

A Graph-based Framework for Coverage Analysis in Autonomous Driving

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Abstract. Coverage analysis is essential for validating the safety of autonomous driving systems, yet existing approaches typically assess coverage factors individually, struggling to capture the complex interactions inherent in traffic scenes. This paper proposes a graph-based framework that represents traffic scenes as hierarchical graphs combining maps with actor relationships, using a two-phase construction algorithm to capture spatial configurations such as leading, following, neighboring, and opposing. We present two complementary coverage methods: a subgraph isomorphism approach that matches scenes against manually defined archetype graphs, and a graph embedding approach using Graph Isomorphism Networks with Edge features (GINE) trained via self-supervised contrastive learning. Validated on real-world Argoverse 2.0 data and synthetic CARLA data, the isomorphism method yields node coverage percentages from predefined archetypes, while the embedding approach reveals latent structure suitable for clustering and anomaly detection. The framework scales efficiently across diverse scenarios without scenario-specific handling and naturally accommodates varying numbers of actors.

Keywords: Coverage Analysis · Autonomous Driving · Traffic Scene Graphs · Subgraph Isomorphism · Graph Neural Networks

1 Introduction

In autonomous driving, coverage analysis is a crucial step to ensure the safety and reliability of the system. At its core, a coverage argument asks whether the scenes observed so far have exhausted the space of traffic scenes that can occur in the real world. This is a completeness argument, not a safety analysis, though any safety analysis can in turn only be carried out on top of such a completeness argument. In most situations, such arguments are collected either per coverage factor, or maybe up to 2 or 3 factor interactions. See for example [7] for a production-grade implementation of state-of-the-art coverage analysis.

In contrast to existing approaches, this paper proposes a graph-based framework for coverage analysis. The framework takes the *complete traffic scene* as the unit of coverage rather than an individual factor: each scene is represented as a

single graph whose nodes carry the parameters describing the individual actors and whose edges carry the parameters describing the relations between them. The key advantage of this representation is that arbitrarily complex scenes—with varying numbers of actors and the higher-order interactions between them—are captured naturally within one object, instead of being decomposed into isolated coverage factors or low-order factor combinations. Coverage is therefore assessed against whole scenes: we ask how completely the observed scene graphs cover those that can occur in reality. While graph-based representations of traffic scenes already exist, they have not been specifically designed for this coverage analysis. This work bridges that gap by utilizing graph-based traffic scene representations explicitly for coverage analysis.

As this paper is focussing on coverage analysis, it is not concerned with the safety of the system, but rather with the completeness of the observed scenes. There are of course many connections between coverage and safety aspects, e.g. a reasonable strategy is to put higher focus in coverage analysis on the scenes that are more safety-critical, however, in this paper the focus is solely on coverage and we leave it to future work to expand this work to include safety considerations.

2 Existing coverage and analysis approaches

Foundational terminology for automated driving is established by the SAE J3016 taxonomy [15], which defines six levels of driving automation and key concepts such as the Dynamic Driving Task (DDT) and Operational Design Domain (ODD). Building on this, [20] provide precise definitions for the commonly conflated terms *scene*, *situation*, and *scenario*, enabling a consistent vocabulary for test design and coverage arguments. Complementing this work, [8] propose object-oriented ontologies for scenario description with explicit linkage to coverage arguments.

Scenario-based testing has emerged as the dominant paradigm for validating automated driving functions. The PEGASUS project [16] introduces a six-layer model to decompose the driving environment into structured logical scenarios, shifting validation from public road testing toward systematic scenario exploration across simulation and field tests. [11] refine this with three abstraction levels—functional, logical, and concrete scenarios—while noting that existing parameter-selection methods lack a systematic way to determine meaningful test coverage. Motivated by this weak point, [17] present trajectory-based clustering of real-world driving data to structure the scenario space and reduce test-set redundancy.

A central challenge is the *approval trap* identified by [21]: statistically demonstrating superior safety through real-world driving alone would require billions of test kilometres, motivating simulation-based tools and systematic coverage methods. General testing concepts [1] are adapted for the domain in [7] through coverage buckets and performance metrics, while ISO 21448 [19] and UL 4600 [10] standardize coverage criteria for automated driving.

More recent work represents traffic scenes directly as graphs to enable scalable, label-free analysis of the scenario space. [23] encode semantic scene graphs into low-dimensional embeddings via a Siamese network to cluster traffic scenes without manual labels, an approach refined through graph-based contrastive learning in [24] and extended to heterogeneous scenario graphs for clustering and similarity search by [14]. Targeting scalability, [13] propose SparScene, a sparse graph-learning framework that exploits lane-graph topology rather than distance thresholds to build structure-aware agent-map connections, scaling scene representation to thousands of agents for large-scale trajectory generation. Cluster-based analyses have likewise been used to assess the completeness of scenario categories [18], complementing the standardized coverage criteria discussed above. These graph-based representations directly motivate our framework, which combines explicit subgraph isomorphism against archetype graphs with learned GINE embeddings to quantify coverage of the scenario space.

3 Defining a traffic scene graph

3.1 Datasets

Argoverse 2.0 [22] is a large-scale dataset of 250,000 scenarios (11 s each) with tracked trajectories for vehicles, pedestrians, and cyclists, plus semantic map information (lane boundaries, crosswalks, traffic signals) across diverse geographic locations. A subset of 17,000 training scenes is used.

CARLA (Car Learning to Act, [4]) is an open-source simulator for autonomous driving research. Version 0.9.15 is used across six maps (Town01–Town05, Town07) with 20–60 vehicles per run, simulating mixed traffic with varied vehicle types and behavioural parameters. The resulting data consists of 2,050 scenes of 11 seconds each; actors spread across the map may form multiple disconnected components per scene.

3.2 Map Graph Construction

The map graph is a directed multigraph $G_{\text{map}} = (V_{\text{map}}, E_{\text{map}})$ where each node $v \in V_{\text{map}}$ represents a lane segment, storing geometric information (boundaries, centerline, length) and semantic attributes (intersection status, road and lane type). Edges encode three spatial relationships between lanes: **following edges** connect lanes forming a continuous path in the same direction, **neighbor edges** connect adjacent lanes traveling in the same direction, and **opposite edges** connect lanes traveling in opposite directions. The map graph is constructed by processing map data from Argoverse or CARLA to extract these relationships. Lanes with geometric overlap are marked as intersection lanes.

3.3 Actor Graph Construction

The actor graph $G_{\text{actor}} = (V_{\text{actor}}, E_{\text{actor}})$ represents dynamic relationships between actors at a specific timestep. Each node $v \in V_{\text{actor}}$ represents an actor and

stores attributes including primary lane ID, all occupied lane IDs, longitudinal position s along the lane centerline, 3D position, longitudinal speed, actor type, and a lane change indicator. Each edge $e \in E_{\text{actor}}$ stores a actor-actor relation type and a path length along the lane network. Although each actor graph is built at a single timestep, the lane change indicator already encodes a one-step temporal signal: it is set by comparing an actor’s lane to its lane one timestep earlier ($\Delta t = 1$ s), so dynamic maneuvers such as cut-in and cut-out are captured even within a single-timestep graph.

Four relationship types are defined between actors, ordered by semantic hierarchy:

1. **Following/Leading:** Longitudinal relationships where actors share the same lane or are connected entirely by following edges.
2. **Neighbor:** Lateral relationships via paths containing exactly one neighbor edge (actors need not be on immediately adjacent lanes).
3. **Opposite:** Relationships via paths containing exactly one opposite edge (actors need not be on immediately opposite lanes).

The attributes and relation types defined above are one example instantiation of the graph model. The parameters of interest, the actor types, and the actor relations are free to be chosen by the user. The coverage methodology is unaffected by these modelling choices.

3.4 Hierarchical Graph Construction Algorithm

The actor graph is constructed in two phases that separate relation discovery from graph building, enabling hierarchical processing and preventing redundant edges. All parameters are listed in Table 1.

Table 1: Input parameters for actor graph construction

Parameter	Type	Description	Value
<i>Distance limits (discovery phase)</i>			
<code>max_distance_lead_veh_m</code>	float	Max distance for leading/following	100
<code>max_distance_neighbor_fwd_m</code>	float	Max distance for forward neighbor	50
<code>max_distance_neighbor_bwd_m</code>	float	Max distance for backward neighbor	50
<code>max_distance_opposite_fwd_m</code>	float	Max distance for forward opposite	100
<code>max_distance_opposite_bwd_m</code>	float	Max distance for backward opposite	10
<i>Node distance limits (construction phase)</i>			
<code>max_node_dist_leading</code>	int	Max edge count in path for lead-3 ing/following	3
<code>max_node_dist_neighbor</code>	int	Max edge count in path for neighbor	2
<code>max_node_dist_opposite</code>	int	Max edge count in path for opposite	2
<i>Timestep configuration</i>			
<code>delta_timestep_s</code>	float	Time step increment in seconds	1.0

Phase 1: Relation Discovery: For each actor pair (A, B) at timestep t , the algorithm determines their primary lanes and checks if a connecting path exists in the map graph. The path structure determines the relationship type: paths consisting entirely of following edges yield following/leading relations, paths with exactly one neighbor (or opposite) edge yield neighbor (or opposite) relations. Both lane-based path length and Euclidean distance are checked against the thresholds in Table 1 to filter distant actors.

Phase 2: Hierarchical Graph Construction: Edges are added in hierarchical order—leading/following first (highest priority), then neighbor, then opposite—each sorted by path length (shortest first). Before adding an edge between actors A and B , a breadth-first search checks whether a path of length $\leq \text{max_node_distance}$ already exists in the current graph. If so, the direct edge is skipped, as the relationship is already implicitly encoded. The graph is updated after each edge addition so that subsequent checks reflect the current state.

This redundancy prevention significantly reduces edges while preserving connectivity. For example, three vehicles in a row need only two lead-follow edges; the third pairwise relation is encoded through the intermediate vehicle. Similarly, for a row of vehicles adjacent to an opposite-direction vehicle, only the closest pairwise relation needs an explicit edge (see Figure 1).

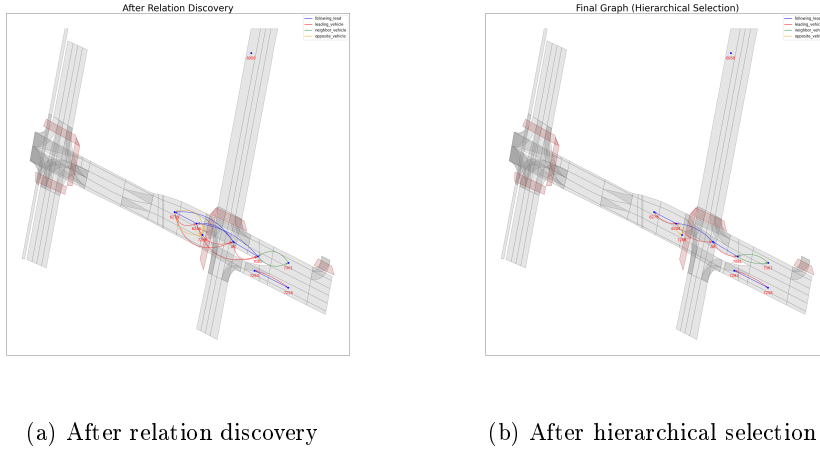


Fig 1: The discovery graph (left) contains all relations within distance limits. Hierarchical selection (right) removes redundant edges representable through existing paths.

4 Create subgraphs for coverage analysis

Traffic scene archetypes from the literature (e.g., lead vehicle following, opposite traffic) can be expressed as small graphs. To determine whether such an archetype appears in a larger traffic scene graph G , we check if any subgraph of G is isomorphic to the archetype graph A . While subgraph isomorphism is NP-hard in general, the graphs considered here are small enough for practical computation using the VF2 algorithm [3].

Our strategy is as follows: (1) Define a set of archetype subgraphs S representing relevant traffic situations. (2) Specify which node and edge attributes are considered for the isomorphism check. (3) For each traffic scene graph G (e.g., from CARLA or Argoverse), check if any subgraph of G is isomorphic to any archetype in S and record the result in a coverage table C .

This bottom-up approach starts from individual situation archetypes and checks their presence across datasets. The resulting coverage table enables follow-up analyses such as visualizing attribute distributions (speed, distance) for matched scenes, cross-tabulating co-occurring archetypes, or computing AV performance metrics conditioned on specific archetypes.

We defined 18 subgraph archetypes covering common traffic scenarios: simple 2-actor patterns (following, opposite, neighbor, used only for isolated pairs), 3-actor patterns (lead vehicle with neighbor, platoons, opposite traffic, lane change/cut-in scenarios with intersection variants), 4-actor patterns (cut-out, multi-vehicle platoons, lead-follow with opposite traffic and intersection variants), and 5-actor patterns combining lead, neighbor, and opposite vehicles with intersection variants. These archetypes are drawn from common scenario primitives in the literature (lead-follow, cut-in, cut-out, opposite traffic, platoons) and constitute a *user-defined input* to the framework rather than a complete or safety-certified taxonomy: practitioners can substitute archetypes specific to their operational design domain or safety case. Completeness of this particular set is therefore not claimed and not required for the method itself. The used archetypes are derived from the experience of the authors and are intended to cover loosely the space of relevant traffic situations with illustrative purposes and is not meant to be exhaustive. In a real safety case, there could be easily several dozens or more archetypes, depending on the level of detail required. One critical aspect in a real application is to assure the completeness of the archetype set.

In a real safety case, the archetypes should be derived top-down from established processes rather than expert intuition: a Hazard Analysis and Risk Assessment (HARA, ISO 26262) yields the hazardous traffic situations to be covered, while UL 4600 requires an explicit argument that the scenario space is sufficiently covered. Each identified hazard maps to an archetype, and any hazard without a matching archetype directly flags a coverage gap and thus a risk of missed hazards.

Crucially, the complementary embedding approach (Section 5) operates *without* predefined archetypes and can surface real-world scenes that match no archetype, indicating candidates for extending the set and thus guarding against

omissions of rare or safety-critical situations. The last step in the analysis of the subgraphs is to identify coverage differences between real-world and simulation datasets. The archetypes provide an intuitive understanding of what traffic situations are missing from the simulated environment.

5 Graph Embeddings for Traffic Scene Analysis

The subgraph isomorphism approach from Section 4 provides a bottom-up methodology for analyzing traffic scenarios through predefined archetypes. However, real-world traffic scene graphs exhibit significantly higher complexity than these patterns, motivating complementary top-down approaches such as graph embeddings.

Embeddings translate raw data into a vector space where distances measure similarity, analogous to the Word2Vec model for words [12]. For traffic scene graphs, embeddings enable coverage analysis by comparing scenes: identifying similar simulation–real-world scenario pairs, detecting near duplicates, or visualizing structures that would be intractable in the original space of all possible traffic scenes.

A Graph Isomorphism Network with Edge features (GINE) [9] is used to generate embeddings, implemented in PyTorch Geometric [6]. GINE is chosen because it allows embedding of edge features, so we may encode actor relation type and distance as integral descriptors for a traffic scene (see Figure 2).

The architecture is shown in Figure 2. Node features are actor type (one-hot: vehicle, pedestrian, cyclist, motorcycle), speed, intersection presence, and lane change since the last timestep. Edge features are edge type (one-hot, 6 spatial relationship categories) and path length between nodes.

The model was trained on CARLA and Argoverse 2.0 using self-supervised contrastive learning [2,9]. For each mini-batch, two augmented views were created by perturbing continuous attributes with Gaussian noise ($\sigma = 0.08$) and randomly dropping edges (probability 0.1). A five-layer GINE encoder produced graph-level representations via concatenated mean, max, and sum pooling, followed by an embedding MLP (ℓ_2 -normalized) and a projection head. The contrastive loss maximized agreement between views of the same graph while contrasting against other in-batch graphs (cosine similarity, $\tau = 0.07$). Optimization used AdamW (weight decay 5×10^{-6}), learning rate warmup over three epochs, and exponential decay (initial rate 0.0015, factor 0.85) across 15 training stages [9,6].

6 Application: Analysing traffic scenes

We apply both methods to CARLA and Argoverse data to identify coverage gaps, targeting not the exact real-world distribution but gaps that could lead to insufficient testing of safety-critical scenarios.

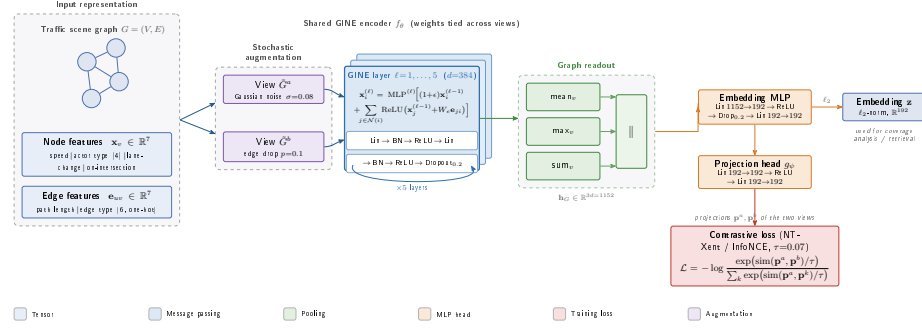


Fig. 2: Architecture of the GINE-based graph embedding model and its self-supervised training. A traffic scene graph G carries 7-dimensional node features (speed, actor type, lane-change and intersection flags) and 7-dimensional edge features (path length and one-hot edge type). Two stochastically augmented views (Gaussian feature noise $\sigma = 0.08$, edge dropout $p = 0.1$) are passed through a weight-shared encoder f_θ of five GINE layers ($d = 384$), each performing edge-conditioned message passing followed by batch normalisation, ReLU and dropout. Node states are aggregated by concatenated mean/max/sum pooling ($\mathbb{R}^{1152}$), mapped by an ℓ_2 -normalised embedding MLP to the scene embedding $z \in \mathbb{R}^{192}$ used downstream, and by a projection head g_ψ to the space where the NT-Xent/InfoNCE contrastive loss ($\tau = 0.07$) is computed between the two views.

6.1 Subgraph Isomorphism Coverage Analysis

We match the 18 archetypes (Table 2) against each scene graph and compare CARLA and Argoverse across two gap dimensions: structural presence of scenario types and parameter distributions within matched scenarios.

Individual Scenario Archetype Coverage Gaps: Figure 3 presents a comprehensive comparison of scenario archetype coverage between CARLA and Argoverse datasets. The dual-axis visualization displays coverage percentages (horizontal bars) alongside coverage differences (diamond markers). Red diamonds indicate scenarios that are more prevalent in Argoverse, while green diamonds mark scenarios with higher CARLA representation.

The coverage distribution reveals a clear pattern: CARLA achieves higher coverage only for the three simple two-actor scenarios (`simple_following`, `simple_neighbor`, `simple_opposite`), as indicated by green diamond markers. For all complex multi-actor scenarios, Argoverse exhibits substantially higher occurrence rates (red diamond markers), demonstrating that the simulation systematically fails to generate complex traffic situations at the same frequency as observed in real-world data.

The identified coverage gaps in CARLA are substantial. The most critical gaps appear in intersection-related archetypes: `cut_out_intersection` shows nearly complete absence in CARLA (less than 1% coverage) while appearing

Table 2: Overview of all subgraph archetypes with their structural properties. Edge types: *fl* = following_lead, *lv* = leading_vehicle, *nv* = neighbor_vehicle, *ov* = opposite_vehicle.

Group	Archetype	Actors	Edges	Edge Types
Simple (2-actor)	Simple Following	2	2	fl, lv
	Simple Opposite	2	2	ov
	Simple Neighbor	2	2	nv
Complex (3-actor)	Lead + Neighbor (intersection)	3	4	fl, lv, nv
	Lead + Neighbor	3	4	fl, lv, nv
	Cut-in	3	4	fl, lv
	Cut-in (intersection)	3	4	fl, lv
	Platoon (intersection)	3	4	fl, lv
	Opposite Traffic (intersection)	3	4	fl, lv, ov
	Lead + Neighbor at Intersection	3	4	fl, lv, nv
	Triple Opposite (intersection)	3	4	ov
	Lead + Following in Back	3	4	fl, lv
Complex (4-actor)	Cut-out	4	6	fl, lv, nv
	Cut-out (intersection)	4	6	fl, lv, nv
	4-Vehicle Platoon (intersection)	4	6	fl, lv
	4-Vehicle Opposite (intersection)	4	6	fl, lv, ov
Complex (5-actor)	Lead + Neighbor + Opposite	5	8	fl, lv, nv, ov
	Lead + Neighbor + Opposite (inters.)	5	8	fl, lv, nv, ov

in approximately 8% of Argoverse scenes. Similarly, `lead_neighbor_opposite_vehicle_intersection` appears in roughly 6% of Argoverse data but is virtually absent in CARLA. The `cut_in_intersection` pattern exhibits a comparable gap of approximately 7 percentage points. Both `cut_in` and `cut_out` scenarios show significant underrepresentation in CARLA regardless of intersections, indicating that the traffic generation script lacks the dynamic lateral maneuvers characteristic of real-world driving.

Parameter Distribution Gaps: Even when a scenario archetype is structurally present in both datasets, continuous parameter distributions can reveal additional coverage gaps. Figure 4 shows role-specific speed distributions within the `lead_vehicle_in_front_with_neighbor_vehicle` scenario.

CARLA concentrates actors at lower speeds (peak around 3–5 m/s) while Argoverse distributions extend to higher speeds (15–20 m/s and beyond). Coverage gaps consistently appear in the 13–17 m/s range for all three roles, demonstrating that coverage gaps exist both at the structural level (which scenarios are present) and at the parameter level within structurally matching scenarios.

The structured, bottom-up approach presented here demonstrates that subgraph isomorphism can effectively characterize traffic scene coverage for predefined archetypes. In a real application, these gaps could now e.g. be checked for safety critical aspects and prioritized accordingly, e.g. by analyzing the distri-

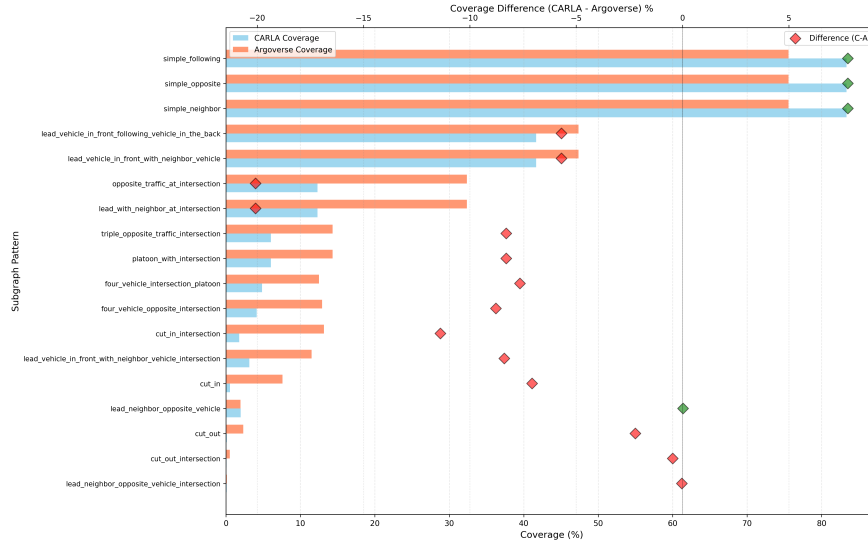


Fig. 3: Scenario archetype coverage comparison between CARLA and Argoverse datasets. Horizontal bars show absolute coverage percentages, while diamond markers indicate coverage differences (CARLA minus Argoverse).

bution of TTC values in the gaps. However, the manual definition of archetypes and the computational cost of isomorphism checking for large scenario collections motivate the graph embedding approach explored in the following paragraph.

6.2 Graph Embeddings Coverage Analysis

The GINE model (Section 5) is trained jointly on CARLA and Argoverse. As a plausibility check, nearest neighbours in embedding space consistently share actor configuration and road geometry with the query (Fig. 5), indicating the model captures scene structure rather than superficial features.

The embedding space is analysed using PCA and t-SNE as shown in Figure 6. The first two principal components already explain 40% of the total variance. Both projections reveal that CARLA and Argoverse scenarios, while sharing a common region corresponding to simple following situations, differ substantially in their density distributions: For the Carla data, the main peak of the distribution is concentrated at higher values of the first principal component, while for Argoverse data the main peak is located at negative values of the first principal component. This indicates that the two datasets differ in their overall scenario characteristics, with CARLA scenarios clustering in a different region of the embedding space than Argoverse scenarios.

Coverage Gap Identification in Embedding Space: No binning is required here, as the full graph is embedded. To quantify coverage gaps, Argoverse graphs whose subgraph isomorphism labels indicate under-represented scenario

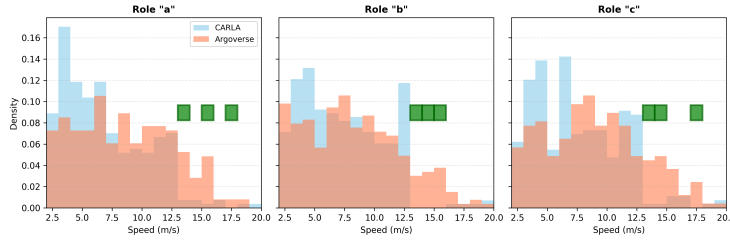


Fig. 4: Role-specific speed distribution comparison for the lead vehicle in front with neighbor vehicle scenario. Only actors with speed ≥ 2 m/s are considered, binned into fixed 1 m/s intervals, and the histograms are density-normalised. Green rectangles mark identified coverage gaps, defined per bin as a normalised Argoverse density ≥ 0.005 together with a CARLA density below 15% of it—i.e. Argoverse shows sufficient density while CARLA is nearly empty. Role “a” = ego vehicle, role “b” = lead vehicle, role “c” = neighbor vehicle.

types are projected into the joint embedding space and compared against the CARLA density. 23,148 Argoverse graphs are identified as coverage gaps, dominated by `cut_in` ($\sim 94\%$ of all Argoverse cut-in instances), `cut_out`, `cut_out_intersection`, and `lead_neighbor_opposite_vehicle_intersection`. These gap graphs cluster in embedding regions where CARLA density is negligible (Fig. 7), reinforcing the gaps identified by subgraph isomorphism.

7 Summary

This paper presented a graph-based framework for coverage analysis in autonomous driving that represents traffic scenes as hierarchical graphs combining map topology with actor relationships. Two complementary analysis approaches were developed: subgraph isomorphism using manually defined archetype graphs for interpretable pattern-based coverage metrics, and graph embeddings via GINE networks trained with contrastive learning for similarity-based assessment, clustering, and anomaly detection.

Both approaches were validated on Argoverse 2.0 and CARLA data. It was shown that hierarchical graphs capture complex traffic situations efficiently and allow the user to define individual specifics on definitions like follow vehicle, opposing vehicle and the set of relevant parameters.

We demonstrated that the graphs capture meaningful traffic scene structure and reveal differences between two datasets (Section 6). First, the bottom-up subgraph approach revealed differences in traffic scenes, combination of traffic scenes and distribution of scenario parameters. Second, the top-down graph embedding approach revealed differences in the embedding space, allowing for further analysis without manual definition of archetypes. This representation can be the foundation for further analysis like similarity search, clustering and

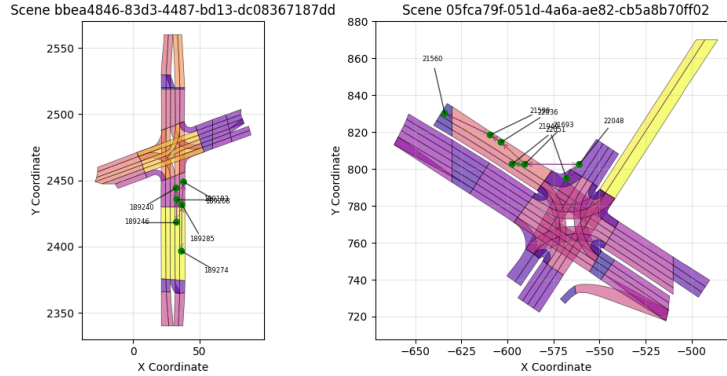


Fig. 5: Plausibility check: for a randomly sampled scenario (left), the most similar scenario by embedding distance is retrieved (right). Retrieved neighbours consistently share the same actor configuration and road geometry, confirming the model captures traffic scene structure.

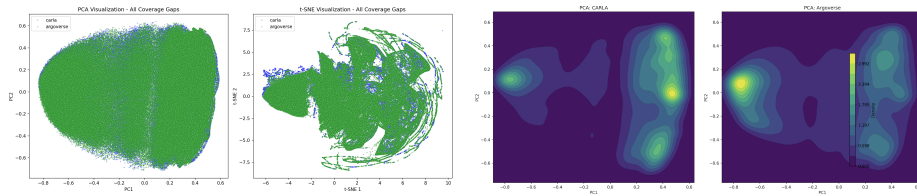


Fig. 6: Left: PCA and t-SNE visualisation of the embedding space for CARLA and Argoverse 2.0 scenarios. Right: PCA kernel density plots for Argoverse 2.0 scenarios, highlighting regions of high real-world density.

anomaly detection. Compared to approaches like TNO Streetwise [5], the embedding approach scales efficiently without scenario-specific handling rules and naturally accommodates varying numbers of actors.

Future work will incorporate temporal information through multi-timestep graph structures, capturing the full dynamics of maneuvers such as cut-in, cut-out, braking, and yielding more faithfully than a single-timestep graph, and close the loop on archetype completeness by automatically extracting new archetype candidates from real-world scenes that populate dense embedding regions yet match no predefined archetype. The source code is publicly available at https://github.com/tmuehlen80/graph_coverage.

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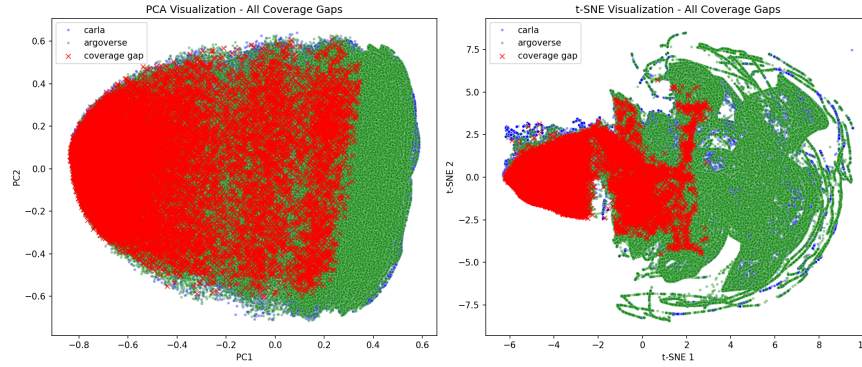


Fig. 7: Coverage gap visualisation in the embedding space. CARLA (blue) and Argoverse (green) scenarios are shown in PCA (left) and t-SNE (right) projections. Red crosses mark the 23,148 Argoverse graphs identified as coverage gaps, concentrated in regions where CARLA density is negligible.

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